

2 quicksort

```
#include <stdio.h>
```

```
void quicksort(int a[], int low, int high) {  
    int p;
```

```
    if (low < high) {
```

```
        p = partition(a, low, high);
```

```
        quicksort(a, low, p-1);
```

```
        quicksort(a, p+1, high);
```

```
    }
```

```
}
```

```
int partition(int a[], int low, int high)  
{
```

```
    int pivot, i, j, temp;
```

```
    pivot = a[low];
```

```
    i = low + 1;
```

```
    j = high;
```

```
    while (i <= j) {
```

```
        while (a[i] <= pivot && i <= high)
```

```
            i++;
```

```
        while (a[j] > pivot)
```

```
            j--;
```

```
        if (i < j) {
```

```
            temp = a[i];
```

```
            a[i] = a[j];
```

```
            a[j] = temp;
```

```
        }
```

```
}
```

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```
temp = a[low];  
a[low] = a[j];  
a[j] = temp;
```

```
return j;
```

```
}
```

```
int main()
```

```
{  
    int a[20], n, i;
```

```
    printf("Enter number of elements:");  
    scanf("%d", &n);
```

```
    printf("Enter elements \n");
```

```
    for(i=0; i<n; i++)
```

```
        scanf("%d", &a[i]);
```

```
}
```

```
    quicksort(a, 0, n-1);
```

```
    printf("Sorted array: \n");
```

```
    for(i=0; i<n; i++)
```

```
        printf("%d", a[i]);
```

```
    return 0;
```

```
}
```


2 insertion sort

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[20], n, i, j, key;
```

```
    printf("Enter number of elements: ");  
    scanf("%d", &n);
```

```
    printf("Enter elements: \n");
```

```
    for(i=0; i<n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    for(i=1; i<n; i++) {
```

```
        key = a[i];
```

```
        j = i - 1;
```

```
        while(j >= 0 && a[j] > key) {
```

```
            a[j+1] = a[j];
```

```
            j--;
```

```
        }
```

```
        a[j+1] = key;
```

```
    }
```

```
    printf("Sorted array: \n");
```

```
    for(i=0; i<n; i++)
```

```
        printf("%d", a[i]);
```

```
    return 0;
```

```
}
```